



Technical Service Bulletin

Technical Service Bulletin: TSB220199

Released Date: 01-Mar-2023

Aftertreatment NOx Sensor Test Available

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- B6.7 CM2350 B121B
- B6.7 CM2450 B155B
- L9 CM2450 L126B
- X12 CM2450 X137B
- X15 CM2350 X114B
- X15 CM2350 X116B
- X15 CM2450 X124B
- X15 CM2450 X134B

Issue Summary

Symptom:

- Fault Codes: 3748, 3749, 5655, 6459 or 6463

Verification

- New Aftertreatment Intake and Outlet NOx Sensor Tests is used when troubleshooting certain engine fault codes. Throughout each test heater response, stability and accuracy are checked to diagnose the NOx sensor.
- The Aftertreatment Intake or Outlet NOx Sensor Test is to be performed when directed by troubleshooting to verify the root cause is identified and repaired the first time without unnecessary repairs.

Resolution

- Perform the Aftertreatment Intake or Outlet NOx Sensor Test when directed by published troubleshooting.
- Reference the appropriate Fault Code Troubleshooting procedure. See corresponding Service Manual. Reference Procedures 014-36 and 014-037 in Section 14.

- The Aftertreatment Intake or Outlet NOx Sensor should **only** be replaced when directed by the Aftertreatment Intake or Outlet NOx Sensor Test. Visual inspection should **not** drive the replacement of NOx Sensor malfunction.

Description of Change

- This document is intended to highlight the ability of the Aftertreatment Intake and Outlet NOx Sensor Tests to diagnose NOx Sensor functionality.
- The Aftertreatment Intake and Outlet NOx Sensor Tests are now available. This diagnostic test was developed to evaluate the functionality of the NOx Sensor.
- The Aftertreatment Intake and Outlet NOx Sensor Test can accurately identify a NOx Sensor malfunction. The use of this diagnostic test reduces the chance of misdiagnosis and is used to validate repair of the NOx Sensor.

Publications Affected

Table 1, Publications Affected

Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	B6.7 CM2350 B121B	5411223	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsyss2/xml/en/procedures/455/455-014-036.html)	14
Service Manual	B6.7 CM2350 B121B	5411223	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsyss2/xml/en/procedures/455/455-014-037.html)	14
Service Manual	B6.7 CM2450 B155B	5613229	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsyss2/xml/en/procedures/647/647-014-036.html)	14

Table 1, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	B6.7 CM2450 B155B	5613229	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/pr ocedures/64 7/647-014-037.html)	14
Service Manual	L9 CM2350 L116B	5411330	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/pr ocedures/45 9/459-014-036.html)	14
Service Manual	L9 CM2350 L116B	5411330	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/pr ocedures/45 9/459-014-037.html)	14
Service Manual	L9 CM2450 L126B	5613077	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/pr ocedures/58 3/583-014-036.html)	14
Service Manual	L9 CM2450 L126B	5613077	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/pr ocedures/58 3/583-014-037.html)	14

Table 1, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	X12 CM2450 X137B	5659705	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/procedures/674/674-014-036.html)	14
Service Manual	X12 CM2450 X137B	5659705	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/procedures/674/674-014-037.html)	14
Service Manual	X15 CM2350 X114B	5411181	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/procedures/377/377-014-036.html)	14
Service Manual	X15 CM2350 X114B	5411181	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/procedures/377/377-014-037.html)	14
Service Manual	X15 CM2350 X116B	5411186	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/procedures/453/453-014-036.html)	14

Table 1, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	X15 CM2350 X116B	5411186	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/procedures/453/453-014-037.html)	14
Service Manual	X15 CM2450 X124B	5504583	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/procedures/564/564-014-036.html)	14
Service Manual	X15 CM2450 X124B	5504583	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/procedures/564/564-014-037.html)	14
Service Manual	X15 CM2450 X134B	5613373	Aftertreatment Outlet NOx Sensor Test	Refer to Procedure 014-036 (/qs3/pubsy s2/xml/en/procedures/643/643-014-036.html)	14
Service Manual	X15 CM2450 X134B	5613373	Aftertreatment Intake NOx Sensor Test	Refer to Procedure 014-037 (/qs3/pubsy s2/xml/en/procedures/643/643-014-037.html)	14

Document History

Date	Details
2022-9-28	Module Created
2022-10-20	Updated Table 1. Added L9 CM2450 L126B.
2023-3-1	Removed L9 CM2350 L116B.

Last Modified: 01-Mar-2023
